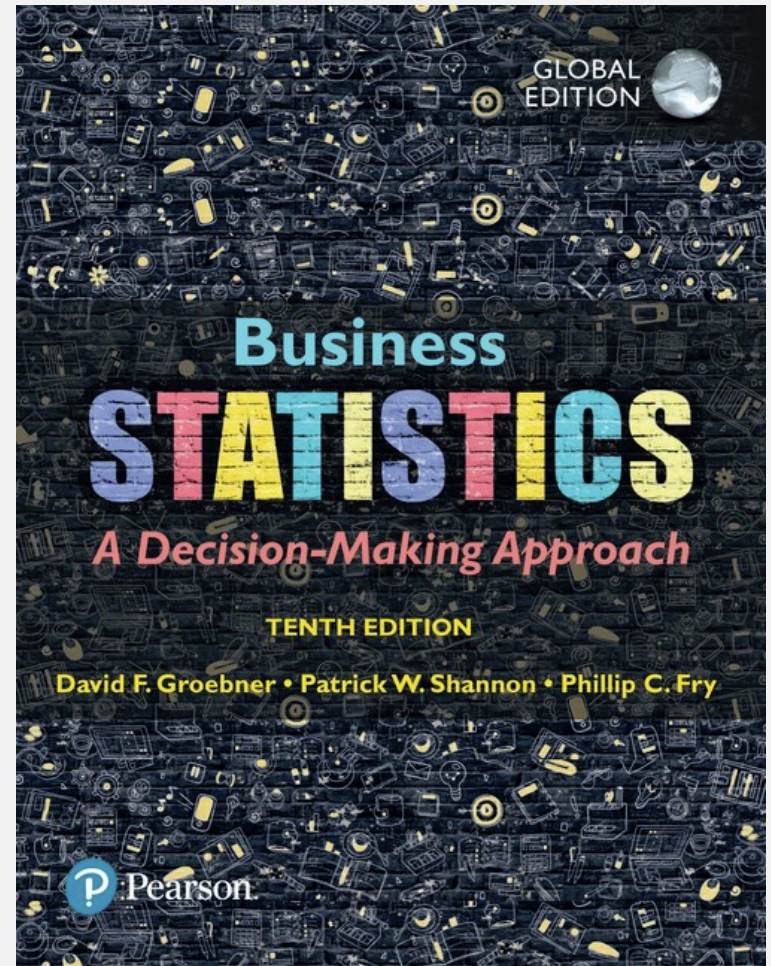


Chapter 8

Estimating Single Population Parameters



Learning Outcomes

- Outcome 1.** Distinguish between a point estimate and a confidence interval estimate.
- Outcome 2.** Construct and interpret a confidence interval estimate for a single population mean using both the standard normal and t distributions.
- Outcome 3.** Determine the required sample size for estimating a single population mean.
- Outcome 4.** Establish and interpret a confidence interval estimate for a single population proportion.
- Outcome 5.** Determine the required sample size for estimating a single population proportion.



8.1 Point and Confidence Interval Estimates for a Population Mean

- **Point Estimate**

- A single statistic, determined from a sample, that is used to estimate the corresponding population parameter

Parameter Point Estimate

	Population	Sample
Mean	μ	$\bar{x}$
Proportion	p	$\bar{p}$

Point estimates are subject to sampling error

Point Estimates

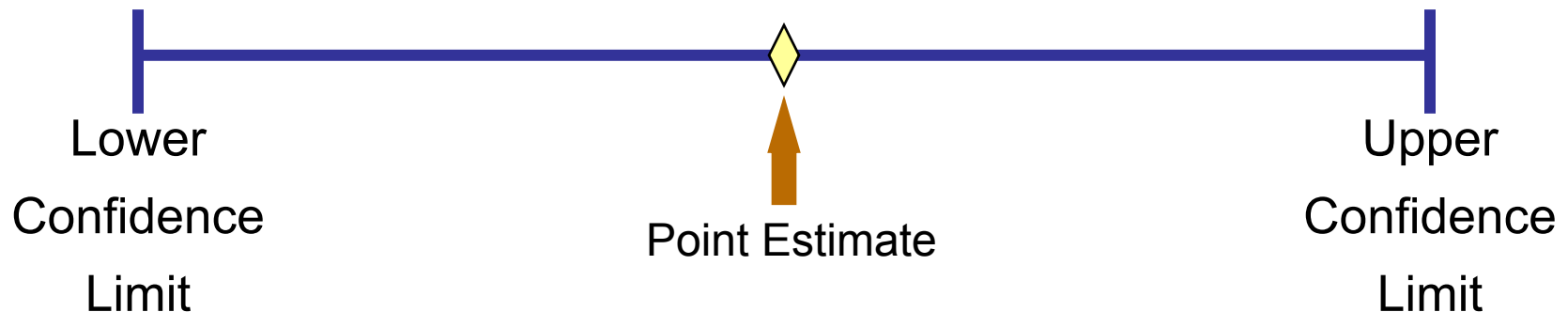
- Population parameter can be estimated with sample statistic (point estimate)

	Population	Sample
Mean		
Proportion		

The Problem With Point Estimates – They are almost always wrong!

Confidence Interval Estimate

- An interval developed from a statistical sample such that if all possible intervals of a given width were constructed, a percentage of these intervals, known as the confidence level, would include the true population parameter



Confidence Interval Estimate – General Format

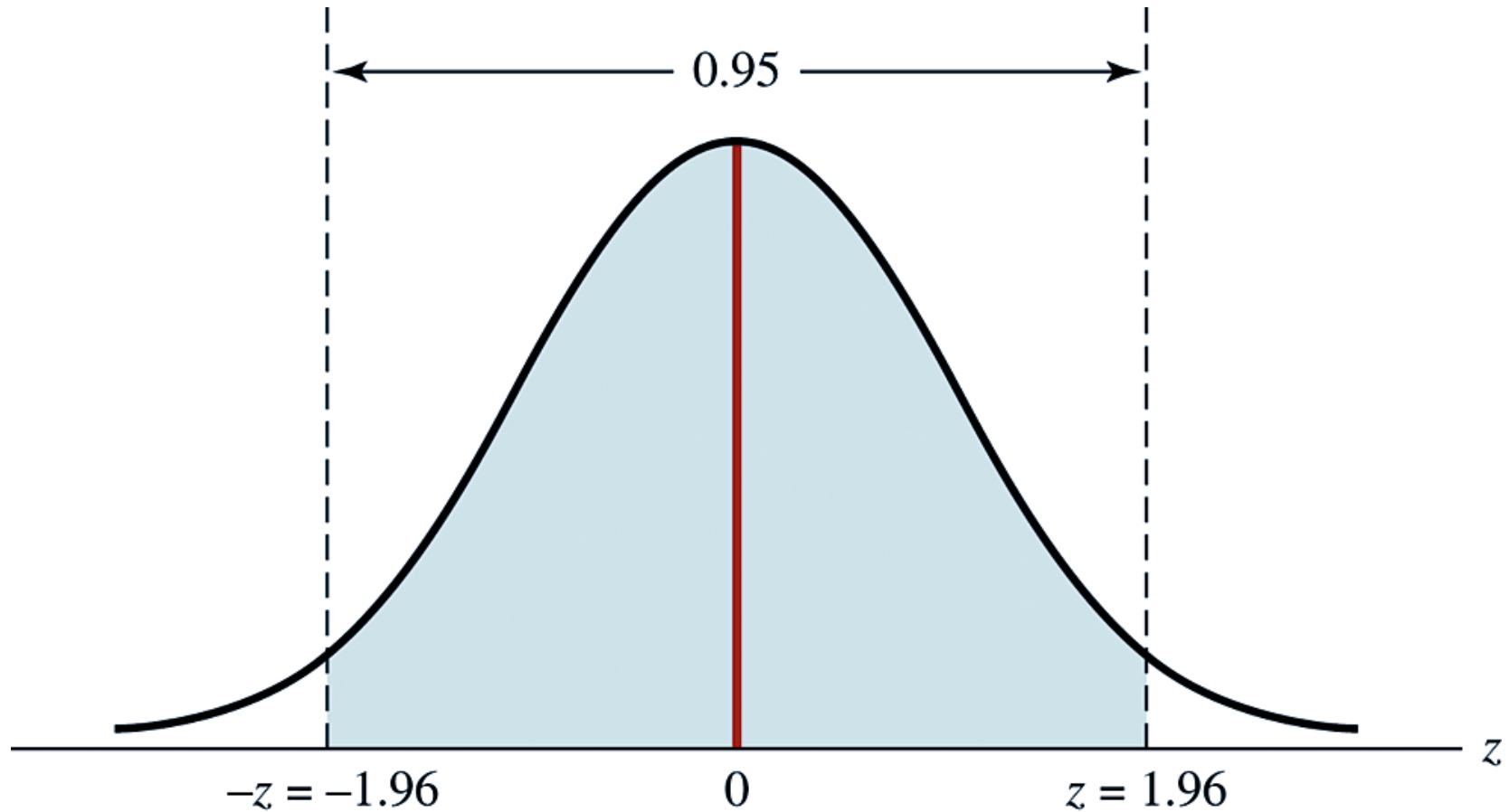
Confidence Interval Estimate

Confidence Level	Critical Value
80%	$Z = 1.28$
90%	$Z = 1.645$
95%	$Z = 1.96$
99%	$Z = 2.575$

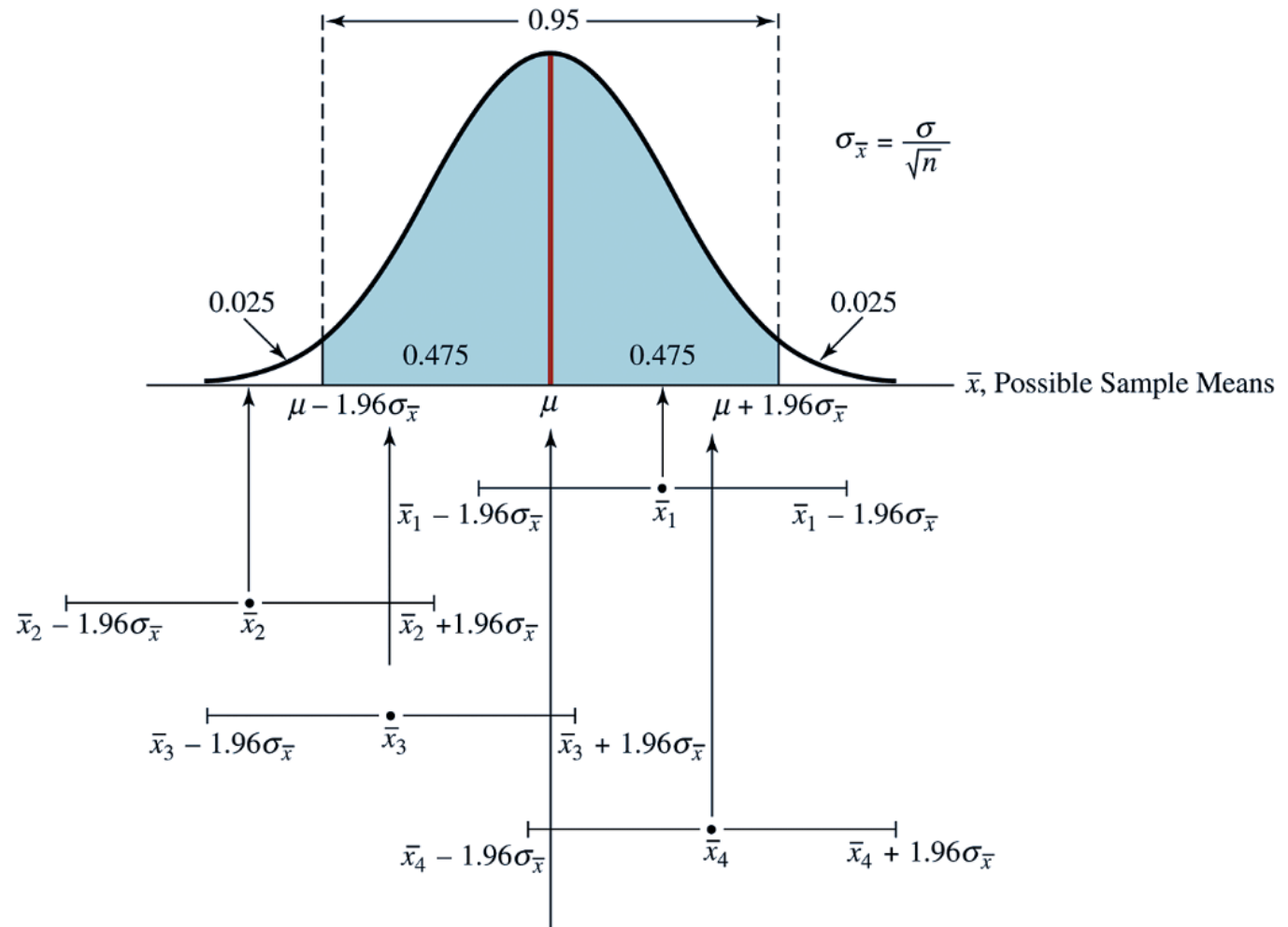
Critical values can be found using the standard normal table, or using Excel's **NORM.S.INV** function



Critical Value – 95% Confidence



95% Confidence Interval



Confidence Interval Estimate - Example

Assuming that the population distribution for credit card balances is normal with a population standard deviation equal to \$400, construct and interpret a 95% confidence interval estimate for the population mean balance if a random sample of $n = 100$ accounts have a sample mean equal to \$2,000.



Confidence Interval Estimate - Example

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Confidence Interval Estimate - Example

Based on the sample data, with 95% confidence, we conclude that the population mean is somewhere between \$1,921.60 and \$2,078.40



Margin of Error- Example

The margin of error indicates that the sample mean of \$2,000 is within \$78.40 of the true population mean.



Confidence Interval Estimate - Example

The distribution of hours worked by students at a university is normally distributed. If the population standard deviation is known to be 5 hours, construct and interpret a 90% confidence interval estimate for the mean hours worked by all students at the university if a random sample of 64 students has a mean equal to 23 hours.



Confidence Interval Estimate - Example

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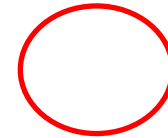
Margin of Error- Example

The margin of error indicates that the sample mean of 23 hours is within 1.03 hours of the true population mean.

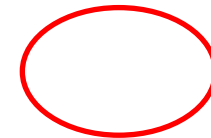


Confidence Level Impact – University Student Work Example

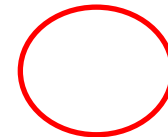
90% Confidence Interval



95% Confidence Interval

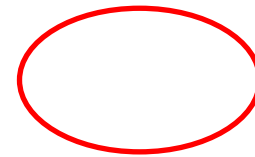


99% Confidence Interval

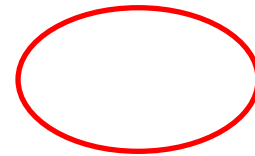


Sample Size Impact – Credit Card Example

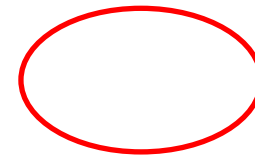
$n = 100$



$n = 200$



$n = 400$



Confidence Interval Objectives

1. Small margin of error
2. High Confidence
3. Small sample size

Conflicting objectives



Conclusions So Far

- Point estimates are subject to sampling error
- For a given sample size, the percentage of possible confidence interval estimates that contain the true parameter is equal to the confidence level
- The margin of error is half the width of the confidence interval estimate
- Increasing the confidence level, increases the margin of error
- Increasing the sample size decreases the margin of error.